Hardware Requirements

To build this project, you will need the following components:

- **ESP32-CAM Module:** With OV2640 camera module.
- **FTDI Programmer (USB to TTL):** Required to upload firmware to the ESP32-CAM (since it lacks a built-in USB port).
- **5V Relay Module (1-Channel):** To act as a switch for the electronic lock.
- **Electronic Door Lock:** E.g., a 12V Solenoid lock or magnetic lock.
- **Power Supply:**
    - 5V / 2A power supply for the ESP32-CAM and Relay.
    - Separate power supply for the lock (e.g., 12V), depending on your lock's specifications.
- **Jumper Wires:** Female-to-Female and Male-to-Female for connections.
- **Breadboard:** (Optional) For easier prototyping and power distribution.

Software Requirements

- **Arduino IDE:** With the ESP32 board manager installed.
- **Python 3.7+:** Installed on your computer/server.